Exp No:11 Experiment to understand pandas library use cases in Data science.
Description: Understand data frame use cases using pandas library

PROGRAM:

```
import pandas as pd

# Create employee dataset

data = {

    "Emp_ID": [101, 102, 103, 104, 105],

    "Name": ["Arun", "Priya", "Kavin", "Divya", "Rahul"],

    "Salary": [45000, 58000, 52000, 75000, 68000]

}

df = pd.DataFrame(data)

# Display dataset

print("Employee Data:")

print(df)

# Display first rows

print("\nFirst Rows:")

print(df.head())

# Display information

print("\nData Information:")

df.info()

# Display salary column

print("\nSalary:")

print(df["Salary"])

# Basic statistics

print("\nSummary Statistics:")

print(df.describe())

# Salary statistics

print("\nAverage Salary:", df["Salary"].mean())

print("Median Salary:", df["Salary"].median())
```

```
print("Maximum Salary:", df["Salary"].max())
print("Minimum Salary:", df["Salary"].min())
```

OUTPUT:

```
Employee Data:
   Emp_ID   Name  Salary
0     101   Arun   45000
1     102  Priya   58000
2     103  Kavin   52000
3     104  Divya   75000
4     105  Rahul   68000

First Rows:
   Emp_ID   Name  Salary
0     101   Arun   45000
1     102  Priya   58000
2     103  Kavin   52000
3     104  Divya   75000
4     105  Rahul   68000

Data Information:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Emp_ID  5 non-null      int64
 1   Name    5 non-null      str
 2   Salary  5 non-null      int64
dtypes: int64(2), str(1)
```

memory usage: 172.0 bytes

Salary:

0 45000

1 58000

2 52000

3 75000

4 68000

Name: Salary, dtype: int64

Summary Statistics:

	Emp_ID	Salary
count	5.000000	5.000000
mean	103.000000	59600.000000
std	1.581139	12054.044964
min	101.000000	45000.000000
25%	102.000000	52000.000000
50%	103.000000	58000.000000
75%	104.000000	68000.000000
max	105.000000	75000.000000

Average Salary: 59600.0

Median Salary: 58000.0

Maximum Salary: 75000

Minimum Salary: 45000